

Supplementary Information

Nuclear nanomedicine using Si nanoparticles as safe and effective carriers of ^{188}Re radionuclide for cancer therapy

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Chromatography data to determine efficiency and stability of Si* NPs -PEG- ^{188}Re complex

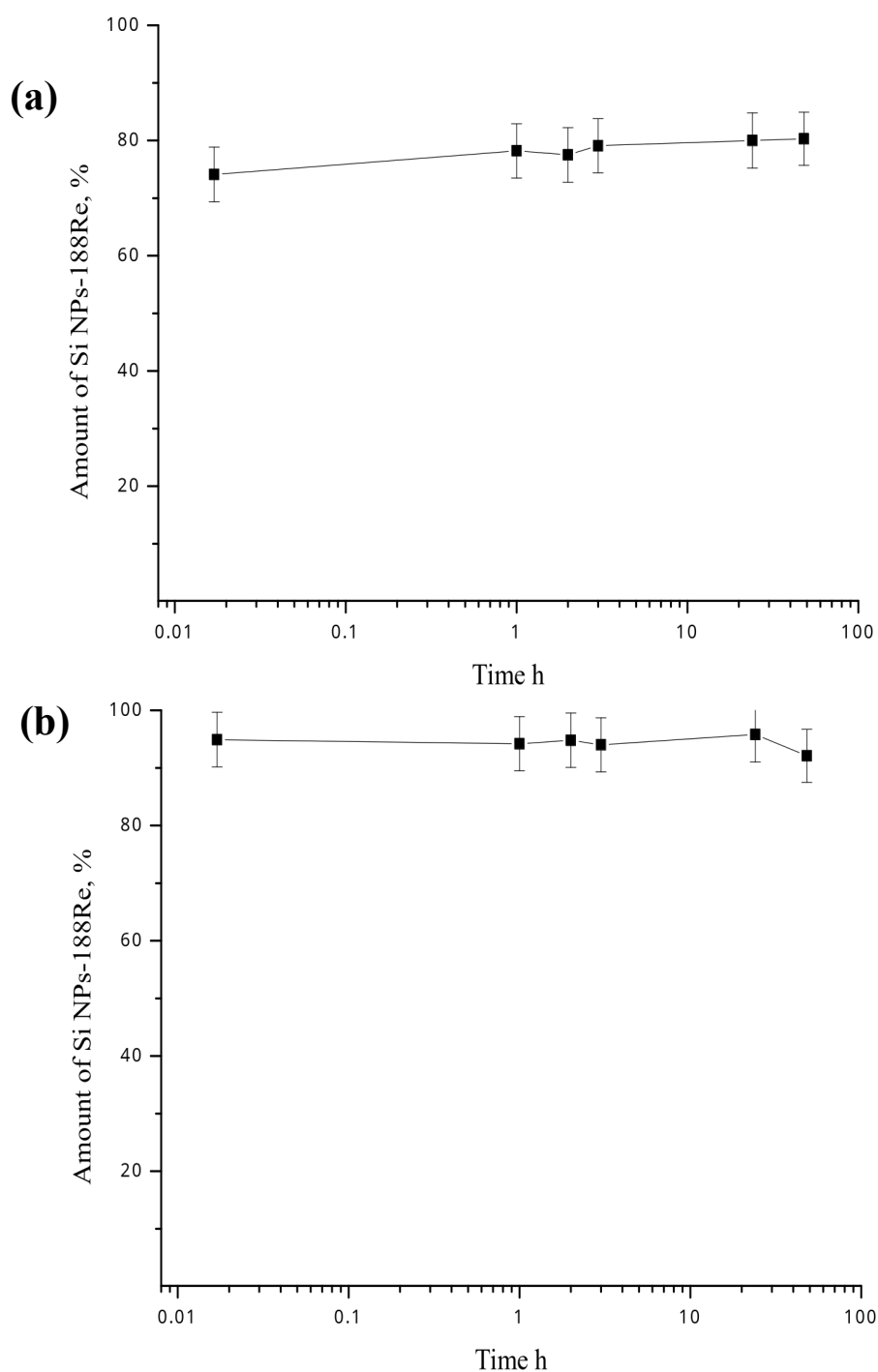


Figure 1S. (a) Dynamics of conjugation of ^{188}Re with the Si*NPs-PEG complex during 48 hours; (b) Temporal stability of the ^{188}Re -based nano-conjugates on Whatman paper after washing out the unconjugated ^{188}Re eluate

As shown in Fig. 1Sa, the efficiency of conjugation is about 80% related to initial radioactivity of ^{188}Re eluate. As follows from Fig. 1Sb, the Si*NPs-PEG- ^{188}Re conjugates were stable during first 48 hours, while the impact of radionuclide impurities after washing out unconjugated ^{188}Re eluate was lower than 5%.